

E-R Model: Relationship Sets, Roles, and Constraints

Visually Structured Notes with Diagrams

Note

Legend. Blue rounded rectangles: *entities*; orange diamonds: *relationships*; yellow ellipses: *attributes*.

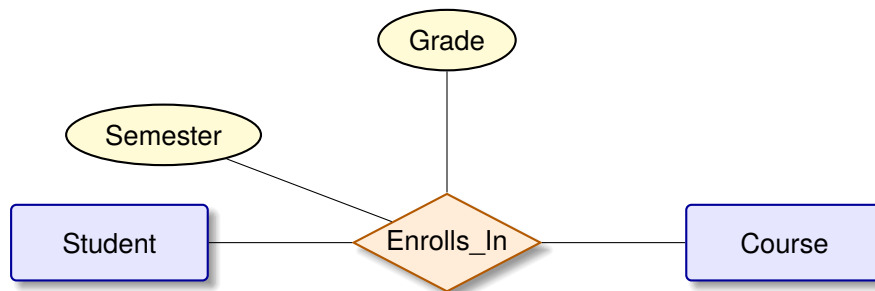
A **directed line (arrow)** denotes “one”, and an **undirected line** denotes “many”.

Double line = total participation.

Relationship Sets with Attributes

Definition

A **relationship set** may have its own *descriptive attributes*, which describe the association between participating entities (not the entities themselves).



Roles in Relationships

Definition

If the same entity set participates multiple times in one relationship, each participation is a **role**. Example: in a *Prerequisite* relationship, the two roles are *course_id* and *prereq_id*.



Constraints in the E-R Model

Definition

An E-R enterprise schema may define constraints to which the database must conform:

1. **Mapping cardinalities (cardinality ratios).**
2. **Participation constraints.**

Cardinality Constraints

Definition

Cardinality constraints specify how many entities in one set can be associated (via a relationship set) with entities in another set. We depict “**one**” with a *directed* line (arrow) and “**many**” with an *undirected* line.

One-to-One (1:1)

Example

A student is associated with at most one instructor via advisor; a student is associated with at most one department via stud_dept.



One-to-Many (1:N)

Example

An instructor is associated with several (possibly 0) students via advisor. A student is associated with at most one instructor via advisor.



Many-to-One (N:1)

Example

A student may be associated with several (including 0) instructors via advisor; each instructor is associated with at most one student via advisor.



Many-to-Many (M:N)

Example

An instructor may advise several (possibly 0) students; a student may have several (possibly 0) instructors (depending on policy).



Participation of an Entity Set in a Relationship Set

Total Participation

Definition

Total participation (double line) means every entity in the set participates in at least one relationship instance.

Example

Participation of Section in Sec_Course is total: every section must have an associated course.



Partial Participation

Definition

Partial participation (single line) means some entities may not participate in any relationship instance.

Example

Participation of Instructor in Advisor is partial: some instructors may advise no students.



Notation for More Complex Constraints (l..h)

Definition

An edge may carry a min..max label where min is the minimum and max the maximum number of relationships an entity instance can participate in along that edge.

1 as minimum $\Rightarrow$ total participation; 1 as maximum $\Rightarrow$ at most one; * as maximum $\Rightarrow$ unbounded.

Example

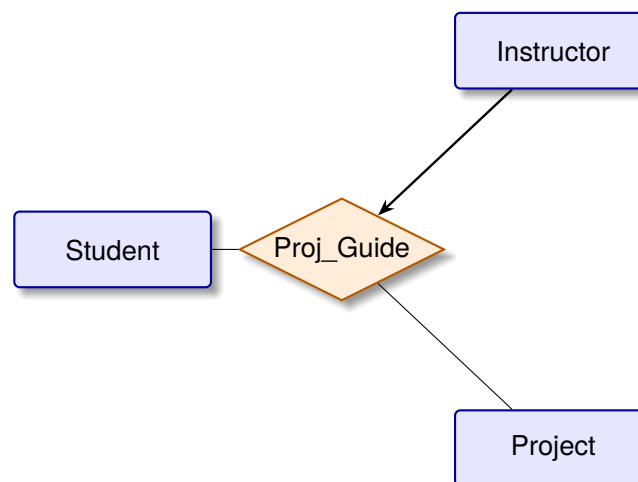
Instructor can advise 0..* students. A Student must have 1..1 advisor.



E-R Diagram with a Ternary Relationship

Definition

For a ternary (or higher-degree) relationship, we allow **at most one arrow** to indicate a cardinality constraint. For example, an arrow to Instructor from Proj_Guide indicates each student has at most one guide for a project.



Note

If more than one arrow is used in a ternary relationship, multiple (conflicting) interpretations are possible (e.g., uniqueness per single entity vs. per pair). To avoid ambiguity, **we forbid using more than one arrow**.

Quick Reference: Mapping Cardinalities

Type	Depiction / Rule
1:1	Arrow on both sides
1:N	Arrow on the "1" side; undirected on the "N" side
N:1	Arrow on the "1" side; undirected on the "N" side
M:N	Undirected on both sides
Total participation	Double line from entity to relationship
Partial participation	Single line from entity to relationship
Min..Max	Edge label 1..h (e.g., 0..*, 1..1)